

Chapter 3 --- SQL Command Categories

SQL can perform many different types of operations. To make these operations easier to understand, SQL commands are divided into five main categories, each with a specific responsibility.

- DDL --- Data Definition Language
- DML --- Data Manipulation Language
- DQL --- Data Query Language
- DCL --- Data Control Language
- TCL --- Transaction Control Language

3.1 DDL --- Data Definition Language

DDL is used to define and manage the structure of a database. It works with databases, tables, columns, data types, constraints, and other database objects.

Common DDL commands:

Command Purpose

CREATE Creates a new database, table, or object
ALTER Changes the structure of an existing table
RENAME Renames a table or column
TRUNCATE Removes all rows from a table, keeping its structure
DROP Permanently deletes a table or database

Simple understanding: DDL creates or changes the structure that will hold the data.

3.2 DML --- Data Manipulation Language

DML is used to manage the records stored inside database tables. It lets you add new records, modify existing records, and remove selected records.

Command Purpose

INSERT Adds new records to a table
UPDATE Modifies existing records
DELETE Removes selected records

Simple understanding: DML works with the data stored inside the structure.

3.3 DQL --- Data Query Language

DQL is used to retrieve information from a database. It lets you view

stored records, search for required information, filter and sort records, calculate summaries, and combine information from different tables.

The main DQL command is SELECT.

```
SELECT column1, column2FROM table_nameWHERE condition;
```

Simple understanding: DQL asks questions about the stored data.

3.4 DCL --- Data Control Language

DCL is used to control access to databases and tables. It determines who can access the database, what data a user can view, what operations a user can perform, and which permissions should be removed.

Command Purpose

GRANT Gives a user specific permissions

REVOKE Removes previously granted permissions

Simple understanding: DCL controls who is allowed to use the data.

3.5 TCL --- Transaction Control Language

TCL is used to manage related database changes as a single transaction. It lets you permanently save changes, cancel changes, create checkpoints during a transaction, and keep multi-step operations consistent.

Command Purpose

COMMIT Permanently saves all changes made in the transaction

ROLLBACK Cancels changes made in the transaction

SAVEPOINT Creates a checkpoint within a transaction to roll back to

Simple understanding: TCL decides whether related changes should be saved or cancelled.

3.6 How the Categories Connect

Imagine you are building a student management database:

Category Responsibility

DDL Creates the database and table structures

DML Adds or changes student records

DQL Retrieves required student information

DCL Controls who can access the information

TCL Saves or cancels related changes

Quick Comparison Table

Category Full Form Main Purpose Key Commands

DDL Data Definition Defines database structure
CREATE, ALTER,
DROP, TRUNCATE,
RENAME

DML Data Manipulation Manages stored records
INSERT, UPDATE,
DELETE

DQL Data Query Retrieves data
SELECT
Language information

DCL Data Control Controls access
GRANT, REVOKE
Language permissions

TCL Transaction Manages transactions
COMMIT, ROLLBACK,
Control Language transactions SAVEPOINT

■ **Final Takeaway** SQL is the language used to work with relational databases. DDL defines the structure, DML manages the records, DQL retrieves the data, DCL controls access, and TCL manages transactions. Each category is learned separately with its own commands and practical examples.

Interview Questions — Chapter 3

Q1. What is the difference between DDL and DML?

Ideal answer: DDL changes database structure; DML changes the records stored inside that structure.

Q2. What is DQL used for?

Ideal answer: DQL retrieves data. SELECT is the main DQL command in the course notes.

Q3. What do DCL commands do?

Ideal answer: GRANT gives permissions and REVOKE removes previously granted permissions.

Q4. What is TCL used for?

Ideal answer: TCL manages transactions using commands such as COMMIT, ROLLBACK, and SAVEPOINT.

Q5. Which category does SELECT belong to?

Ideal answer: DQL in the classification used in these notes.